

University Timetable & Enrollment Portal

User Manual

Md. Fuyad Hassan

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Abstract

This document serves as the official User Manual for the University Timetable & Enrollment Portal. It covers system requirements, setup instructions, graphical user interface (GUI) details, step-by-step usage procedures, database configuration, and troubleshooting steps.

Introduction

The **University Timetable & Enrollment Portal** is a lightweight desktop-based system built using Java Swing and SQLite. It provides an efficient platform for academic departments to register students, define course units, and create weekly class schedules (timetables).

Prerequisites and Setup

System Requirements

- **Operating System:** Windows 10 or 11
- **Java Development Kit (JDK):** JDK 22 or higher (to compile from source)
- **Database:** SQLite JDBC driver (bundled within the project)

Folder Hierarchy

Ensure the following directory structure is maintained:

- `src/dbms/` -- Java source files.
- `lib/` -- Dependency JARs containing `sqlite-jdbc-3.42.0.0.jar`.
- `bin/` -- Destination for compiled class files.
- `timetable.db` -- The database file storing tables for students, units, and schedules.

- `build.bat` -- Compilation and app packaging script.

Building the Project

To build the application, execute the provided batch script in the terminal:

```
[language=bash]  
build.bat
```

This will clean previous builds, compile the Java files, bundle them into a JAR, and run `jpackage` to generate a standalone Windows application under `build-dist/`.

System Architecture

The system is constructed using standard Object-Oriented Design patterns:

DatabaseManager

- Connects to the SQLite database via JDBC and initializes the database tables (`students`, `units`, `timetable`) with foreign keys and cascade rules.

Entity Models

- Object representations of entities (`Student`, `Unit`, `Timetable`) mapped to database rows.

GUI Controller

- `MainDashboard.java` wraps Swing window components, sets the system Look & Feel, and binds action listeners to handle CRUD operations.

User Interface Guide

The interface is split into three main tabs:

Manage Students Tab

This tab manages student records.

1. **Inputs:** Student ID (integer), Name, Email, Gender (Male, Female, Other), and Enrollment Date.

2. **Select Date:** Launches a custom date picker popup. Use the calendar to select a date and click **OK**.
3. **Add Student:** Saves the student profile to the database.
4. **Update Student:** Edits a student's email, name, gender, or date based on their ID.
5. **Delete Student:** Deletes a student profile and automatically clears any timetable entries mapped to them.

Manage Units Tab

This tab manages course units.

1. **Inputs:** Unit Code (unique string), Unit Title, Credits (1--4), and Department.
2. **Add Unit:** Inserts a new unit into the database.
3. **Update Unit / Delete Unit:** Used to edit or delete courses. Deleting a unit automatically removes its schedules.

Manage Schedules Tab

This tab schedules classes by linking students to course units.

1. **Inputs:** Select Student (dropdown), Select Unit (dropdown), Class Day (Monday--Sunday), and Class Time.
2. **Add Schedule:** Creates a new class allocation.
3. **Delete Schedule:** Removes a selected schedule slot.

Troubleshooting

Missing SQLite Driver

-- If you see a class loading error, verify that `sqlite-jdbc-3.42.0.0.jar` is present in the `lib/` directory.

Foreign Key Failures

-- Ensure that when scheduling a class, both the student and unit are registered first.

Database Locked

-- Close any SQLite database editors before running the application to avoid file lockups.